



Machine Learning 3xx

Quiz - 1 (15 Marks) (Duration 60 min*)



Beginners (8 marks - 1 mark each)

- Maximum A Posteriori (MAP) estimation differs from Maximum Likelihood Estimation (MLE) primarily because MAP incorporates which of the following?
A. A non-probabilistic L1 penalty B. A prior distribution over the model parameters C. Empirical risk minimization D. Linear model
B
- The logit function maps probabilities in $(0, 1)$ to real numbers
A. $(-1, 1)$ B. $(0, \infty)$ C. $(-\infty, \infty)$ D. $[0, 1]$
C
- You observe 5 events occurring in a fixed time window modeled by a Poisson distribution with parameter λ , where $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$. What is the Maximum Likelihood Estimate (MLE) for λ ?
A. .5 B. .2 C. 5! D. 5
D
- A binary model is evaluated on 100 samples and produces the following results: $TP = 40$, $FP = 10$, $FN = 10$, $TN = 40$. What are the model's Precision and Recall?
0.8, 0.8
- For a logistic regression model defined by $P(Y = 1|X) = \sigma(w^T x + b)$, what condition on $w^T x + b$ determines the decision boundary when using a default probability threshold of 0.5?
The decision boundary is defined by $w^T x + b = 0$. Since the sigmoid activation is $\sigma(z) = \frac{1}{1+e^{-z}}$, setting $\sigma(w^T x + b) = 0.5$ solves directly to $w^T x + b = 0$.
- Which statement is TRUE?
A. None of the below B. As the number of parameters p increases while sample size N is held fixed, the model's training loss generally decreases or stays the same, even if the new parameters represent pure noise. C. Holding the dataset size N fixed, increasing the number of model parameters p monotonically increases model bias while decreasing model variance. D. For a fixed dataset size N , as the number of estimated parameters p increases, the influence of the prior distribution $P(\theta)$ in MAP estimation becomes weaker relative to the likelihood.
B
- The False Positive Rate (FPR) plotted on the x-axis of an ROC curve can be calculated using which relationship with Specificity
A. 1 - Precision B. 1 - Recall C. 1 - Specificity D. 1/Precision
C
- Changing the probability threshold of a logistic regression classifier from 0.5 to 0.8 will alter the Area Under the Receiver Operating Characteristic curve (ROC-AUC). True/False?
False

Intermediate (4 marks - 1 mark each)

- In Ridge Regression, as the regularization hyperparameter $\lambda \rightarrow \infty$, what happens to the model's **bias** and **variance**?
As $\lambda \rightarrow \infty$, the weight vector $w \rightarrow 0$. The model becomes an extremely simple constant predictor (high bias/underfitting), but its predictions do not vary across different training samples (zero variance).
- When the number of parameters/features p exceeds the number of sample observations N ($p > N$), the Ordinary Least Squares (OLS) closed-form solution yields a unique global solution. True/False & Why?
When $p > N$, the design matrix X (dimension $N \times p$) has a maximum rank of N . Consequently, the $p \times p$ matrix $X^T X$ is rank-deficient and singular (non-invertible). There are infinitely many weight vectors w that achieve zero training error, requiring regularization.

3. An airport security team deploys an AI classifier on luggage X-ray scans to identify dangerous items (weapons/explosives). Out of 100,000 bags scanned daily, on average 20 contain hazardous items. Model A achieves 99.9% accuracy, identifying 10 hazards correctly while missing 10 hazards (False Negatives), and producing 90 false alarms (False Positives). Model B achieves 98.2% accuracy, identifying all 20 hazards correctly (0 False Negatives), but producing 1,800 false alarms (False Positives). Which model should the airport security team select? Which evaluation metric should be prioritized, and how can the operating threshold of the chosen classifier be adjusted to optimize safety?

In high-stakes threat detection, the cost of a False Negative (a weapon slipping onto a plane) is catastrophic, whereas the cost of a False Positive (a manual bag inspection) is merely a minor delay. Recall, which must be maximized. Lower the probability threshold τ (e.g., from 0.50 down to 0.05). This forces the model to classify even borderline scans as positive, driving False Negatives to zero at the expense of lower precision.

4. A hospital wants to predict the number of daily emergency room admissions ($y \in \{0, 1, 2, 3, \dots\}$) based on weather factors and seasonal data. A standard linear regression model predicts -4.2 admissions on warm, low-traffic weekdays. Additionally, the variance of the prediction errors increases as the average number of admissions increases. Why is standard linear regression failing here? How do we fix this?

Standard linear regression fails because it assumes the output can be any real number and that its errors follow a normal (Gaussian) distribution. OLS assumes continuous target variables on $(-\infty, \infty)$ and constant error variance (homoscedasticity). Count data cannot be negative and often follows a different distribution, like a Poisson distribution, hence we require a GLM.

Advanced (3 marks)

1. In Generalized Linear Models (GLMs), the response variable Y belongs to the exponential dispersion family, whose probability density/mass function can be expressed in the form:

$$\log f(y; \theta, \phi) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

It can be shown that the mean is given by $E[Y] = \mu = b'(\theta)$ and the variance is given by $\text{Var}(Y) = b''(\theta)a(\phi) = V(\mu)a(\phi)$, where $V(\mu)$ is the variance function & $b(\theta)$ the cumulant function. Also, $a(\phi) = \frac{\phi}{w}$ is a strictly pre-defined scale function, where ϕ handles unmeasured scale variation & w represents optional a priori weights.

Find $\text{Var}(Y)$ & $V(\mu)$ for Logistic Regression?

Hint: The variance of a binary choice is mathematically locked to its mean probability. There is no room for a free-floating variance scaling factor.

Bonus (1 + 1 marks)

1. A consistent estimator has the property that as the number of data points (which are used to estimate the parameters) increases, i.e. $N \rightarrow \infty$, the estimates converge in probability to the true parameters. Consider the Maximum Likelihood Estimator (MLE) for the variance parameter σ^2 of a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ with unknown mean μ :

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$$

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- (a) Derive the expectation $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2]$ and prove that $\hat{\sigma}_{\text{MLE}}^2$ is a biased estimator for any finite N .
 (b) Prove that $\hat{\sigma}_{\text{MLE}}^2$ is nevertheless **asymptotically unbiased** and **consistent** as $N \rightarrow \infty$.

2. +1 for scoring ≥ 13 marks in the exam without bonus.